

Question	Marking Guidance	Mark	Comments
08.1	1. Antibiotic use is common/regular (in hospitals) OR (Resistant) bacteria easily/readily/often spread (in hospitals); 2. Bacteria with resistant <u>allele</u> (survive), reproduce and pass on <u>allele</u> ; 3. (So resistant) <u>allele</u> frequency increases OR (So) number of bacteria with (resistant) <u>allele</u> increases;	3 (3 x AO1)	2. and 3. Reject immunity for resistance only once Ignore reference to mutation 2 and 3. Penalise gene for allele once only 2. Accept divide/replicate OR binary fission for reproduce

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08.2	Correct answer of $-0.9 = 2$ marks;; 228 (correct numerator) = 1 mark OR 120 (correct denominator) = 1 mark OR 1.9 (correct calculated quotient) = 1 mark OR 0.9 (correct answer but positive integer) = 1 mark ;	2 (2 x AO2)	Accept for 2 marks, - 0.9 as a fraction

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08.3	Strong/significant negative correlation;	1 (1 x AO3)	

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08.4	Correct answer of 8910;; 3510 (correct difference between total number of plasmid kbp and non-plasmid kbp) = 1 mark OR x1.65 (correct multiplication factor for percentage change) = 1 mark OR Incorrect answer, but correct answer shown in working = 1 mark;	2 (2 x AO3)	

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08.5	Germination phase 1. No DNA replication OR No plasmid (B) replication OR No binary fission; Growth phase 2. Plasmid (B) replication occurs OR (Rate of) plasmid (B) replication increases OR Binary fission (only) occurs; Dormant phase 3. Plasmid (B) replication occurs and no binary fission;	3 max (3 x AO3)	Each mark point must be clearly linked to the correct phase. Accept correct time references to indicate each phase Accept references to numbers to identify each phase, eg first phase for 'germination phase' Accept 'cell division' for 'binary fission' Ignore 'cloudiness' and 'number of cells'
